Case A

Mrs. Gupta — 50-Year-Old Graphic Designer | Working Diagnosis: Cervical Postural Dysfunction / Upper Crossed Syndrome

Target: 70%+ | Musculoskeletal (MSK) Physiotherapy

■ PROBLEM LIST

- Dull, persistent cervical and suboccipital pain (1 year)
- Reduced end-range cervical extension
- Protracted head posture + rounded shoulders
- Tightness of upper trapezius and levator scapulae (bilateral)
- Weakness of deep cervical flexors and lower scapular stabilisers (presumed)
- Known osteoporosis — bone fragility risk
- Difficulty sustaining prolonged screen-based work
- Heaviness at skull base limiting concentration/productivity
- Likely sleep disruption secondary to discomfort
- Potential limitation in driving (cervical rotation for shoulder-checking)

■ RED FLAGS

- Bilateral upper limb paraesthesia / hand clumsiness / gait disturbance
- Unexplained weight loss, night sweats, night pain unrelieved by position change
- Sudden onset of severe headache ("worst headache of my life")
- History of significant fall or trauma in context of osteoporosis
- Dysphagia, hoarseness, or cranial nerve signs

■ PATHOPHYSIOLOGY

- This creates a "crossed" pattern of tightness and weakness across the shoulder girdle and cervical spine.
- The protracted head posture shifts the centre of gravity of the head (approx.
- 5 kg) anterior to the cervical spine, increasing compressive forces and erector muscle demand by approximately 4–5x per inch of forward displacement.

■ DIFFERENTIAL DIAGNOSES

1. 1. Cervical Spondylosis (Degenerative Disc/Facet Joint Disease)

- Age-appropriate degenerative change at 50 years
- Neck stiffness, particularly end-range extension/rotation
- Deep aching pain — can mimic postural pain

2. 2. Cervical Radiculopathy (most likely C5–C6)

- Dermatomal radiation into upper limb
- Positive Spurling's test (positive in this presentation — ABSENT here)
- Myotomal weakness, reflex changes

■ OBJECTIVE ASSESSMENTS & HOW TO PERFORM

Cranio-Cervical Flexion Test (CCFT)

What it Measures: Endurance and segmental activation of the deep cervical flexors (DCF) — longus colli and longus capitis — which are the key stabilisers inhibited in Upper Cross | How to Perform: 1. Patient lies supine with a small pillow to maintain neutral cervical position. 2

Spurling's Test (Foraminal Compression Test)

What it Measures: Cervical nerve root compression — narrows the intervertebral foramen to reproduce radicular symptoms. | How to Perform: 1. Patient seated. 2. Passively laterally flex and rotate the cervical spine toward the symptomatic side. 3. Apply gentle axial compression (a

Deep Neck Flexor Endurance Test (Head Lift Test)

What it Measures: Endurance capacity of the deep cervical flexor group — a functional outcome marker for cervical stability and rehabilitation progress. | How to Perform: 1. Patient supine. 2. Instruct: "Tuck your chin gently and lift your head 2.5 cm off the bed." 3. Time how lo

■ TREATMENT PLAN

■ Initial Phase — Weeks 1–4 (Pain Relief, Activation, Posture Correction)

- Postural correction at the workstation: "Monitor should be at eye level. Your ears should be directly above your shoulders. Set a timer to stand, stretch, and reset your posture every 30–40 minutes — micro-breaks are highly effective."
- Pain is mechanical, not structural damage: "Your scan (if done) or symptoms suggest this is a postural and muscular issue, not damage to your discs or nerves. Movement and activity are safe and will help — resting too much will make it worse."
- Osteoporosis and exercise safety: "Because of your osteoporosis, we will avoid any heavy loading or manipulation of the neck. All exercises I give you are safe and will actually help protect your bones through gentle loading."
- Sleep hygiene and pillow advice: "Use a contoured cervical pillow that supports the natural neck curve. Avoid sleeping in a twisted position or on your front. Side-lying with one pillow is ideal."

■ Progression Criteria (Before Advancing to Phase 2)

- NPRS ≤3/10 at rest and during workstation activities (from baseline)
- CCFT sustained at ≥24 mmHg without SCM substitution for 10 seconds

• Deep neck flexor endurance hold ≥30 seconds
• Full pain-free cervical extension ROM (re-measured with goniometer)
• Patient reports improved ability to sustain 2+ hours at workstation (PSFS)

■ Long-Term Plan — Weeks 4–12 (Strength, Function & Self-Management)

• Progressive strengthening: Advance to resistance band rows, prone cobra, wall angels (shoulder mobility), and DCF loaded holds. Gradual progression of sets/reps/resistance aligned with bone loading guidelines for osteoporosis.
• Postural retraining: Introduce dynamic postural correction during task-specific activities (e.g., practising sustained computer work in upright posture, gradually increasing tolerance from 30 min to 2 hours).
• Self-management programme: Written home exercise programme with clear instructions and photo guides. Discuss return-to-activity milestones and signs to watch for (neurological symptoms = return to physiotherapist immediately).
• Workstation ergonomic assessment: Provide written ergonomic checklist or refer for formal ergonomic assessment of her graphic design workstation.
• Discharge criteria: NPRS ≤2/10 at rest and during sustained work; NDI score <10% (minimal disability); independent with full home exercise programme; full cervical ROM in all planes.